

The Effect of Trees on Urban Microclimate and Outdoor Thermal Comfort in the City of Beirut: A Model-Based Study

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ABSTRACT HEADING

Urban heat island (UHI) is a common problem marring urban cities around the world due to its impact on humans and the environment. The underlying challenge arises from the interplay of the contributing factors such as the building's density, the surface albedo, the thermal energy storage capacity of the material, the green spaces, and the anthropogenic heat. This research aims to identify and assess urban vegetation scenarios to mitigate the UHI effect in a neighborhood in Beirut city, the capital of Lebanon. To this end, we employ the ENVI-met software V4.4 to predict the spatio-temporal distribution of the relevant thermodynamic variables and to compare the base case scenario with the urban vegetation scenario. Simulations, conducted on a summer day, are carried on a $200 \times 200 \times 46$ m domain at a resolution of 2 meters, which allows the study of small-scale interfaces between individual buildings, surfaces, and vegetation. The model considers the physical environment such as trees and buildings, the surface characteristics such as roof and pavements, and the material and surface properties. These were selected based on a survey conducted in the chosen area. The model also accounts for solar radiation based on the date, time, geographic coordinates, and coverage of the cloud for the studied environment. The initial input for the software will be the air temperature, the wind speed, the wind direction, the relative humidity, in addition to the boundary condition. Assessment of the proposed mitigation strategy is based on comparing with the base case scenario the following four variables: air temperature, mean radiant temperature, physiological equivalent temperature (PET), and predicted mean vote (PMV).

INTRODUCTION

Urban heat island (UHI) is a common problem in urban cities around the world. The term urban heat island is related to higher urban temperatures in city centers compared to the suburban or surrounding rural areas [1]. This phenomenon results from the density of buildings that are characterized by low albedo and high thermal energy storage capacity, in addition to the reduction of the green spaces and the production of the anthropogenic heat [2]. The rapid spread of humans and the increase in population, in addition to the expansion of buildings in the urban cities, influence the urban microclimate and the outdoor conditions of human comfort. Urban microclimate implies

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the variation in temperature, wind, humidity, and solar radiation, which are impacted by urban morphology parameters such as the building infrastructure, surface materials, and vegetation. The limited open spaces, the lack of vegetation, and the complexity of the urban environment can lead to urban environmental decay in addition to other effects such as poor air quality and traffic congestion. Greening the cities could be a way to mitigate the UHI and decrease the outdoor temperature during summer days. Increasing vegetation plays a vital role in cooling the urban environment through the evapotranspiration process and the shading and solar control. Moreover, the trees are distinguished by their low heat capacity, thus less heat accumulation in green spaces [3]. Another element, which is the sky view factor (SVF), is an essential factor in morphology urban microclimate studies [4]. Adding vegetation in open spaces will lead to a decrease in the SVF, thus decreasing the air temperature. In an urban area with high SVF, the air temperature was higher in the daytime due to the high heat capacity of urban materials in open spaces [5].

Therefore, this study aims to find out the effect of urban vegetation in outdoor spaces for reducing air temperature, enhancing thermal comfort, and mitigating urban heat island.

METHODOLOGY

Lebanon is a small country in the Middle East located on the eastern coast of the Mediterranean Sea, with a surface area of 10,452 km² and a total population of 5,988,000 [6]. Its climate is characterized by a hot, humid summer and a mild winter. Recent data shows that 87.8% of this population lives in urban areas [6].

The study will be conducted in a neighborhood of Beirut, which is called Hamra, as shown in Figure 1, with an area of 200 * 200 m². This area was chosen as a representative of dense built-up neighborhoods with very low tree fraction (1%) and considerable variability of building heights. The focus is to find the effect of urban vegetation in mitigating the UHI effect in Beirut, thus improving the thermal performance in outdoor spaces.

The base case and the vegetation scenario were examined through simulations using the numerical three-dimensional microclimate model, ENVI-met V4.4. The outputs were compared with the base case scenario through air temperature, mean radiant temperature, Physiologically Equivalent Temperature (PET), and the Predicted Mean Vote (PMV) to find the effect of the urban vegetation scenario in mitigating urban heat island.



Figure 1. The geographic location of Beirut and the studied area [7].

Base Case Scenario

As shown in Figure 2, the selected area and its surroundings were digitized on the software after it was traced from the satellite view (Google maps) Figure 1, and then defined through the CAD map. The base case scenario selected for the study was represented with 100 x 100 grids of 2 m resolution. After a survey was conducted in the area, the buildings had different materials according to their age, and variable heights. The ground surface is mostly covered by asphalt, except the surfaces allocated for the pedestrians.

The simulation was conducted on August 20th, a hot summer day, starting from 7:00 am and lasting for 24 hours. We study the effect, during the day, of the urban vegetation on two streets (S1 and S2) of different orientations. The initial meteorological conditions, which correspond to the actual conditions, are represented in Table 1. Table 2 lists the type and the total number of trees in the studied area in addition to their height and foliage albedo that controls the solar radiation incidence on the surface standing below it.



Figure 2. Base case scenario with two different streets.

Table 1. Initial Parameters for the Simulation

Input	
Starting Time	7:00 AM (for 24 hrs.)
Air Temperature	30 °C
Wind Speed	1.3 m/sec
Wind Direction	220 (South-West)
Relative Humidity	65 %
Lateral Boundary Condition	Cyclic
Building Interior Temp.	21 °C

Table 2. Type and Total Number of Trees in the Studied Area

Type of Trees	Height of tree (m)	Foliage Albedo (α)	Quantity
Ficus Elastica	18	0.31	1
Ailanthus Altissima	18	0.28	1
Brachychiton Discolor	10	0.2	5
Syagrus Romanzoffiana	10	0.15	1
Eucalyptus Sp.	8	0.27	4
Ficus Retusa	7	0.27	1
Ligustrum Lucidum	7	0.2	1
Ficus Benjamina	6	0.28	1

Vegetation Scenario

Vegetation can mitigate extreme temperatures and provide better conditions of thermal comfort, as already mentioned above. We explore the impact on the comfort of adding a 25% tree fraction of Ficus Elastica to the current vegetation and trees, as shown in Figure 3. In the proposed scenario, the Ficus Elastica, which is a broadleaf evergreen tree, is planted across the sidewalks and pedestrian streets to provide enough shade and high foliage albedo. All other parameters and conditions remain the same as in the base case. The trees were located in a way to take advantage of their type and use to mitigate the urban heat stress and thermal discomfort.

Our choice of Ficus Elastica is in accordance with the principle of climactic design [8], which states that evergreen trees should be planted on the North-side for wind protection in winter and cooling purposes during summer.

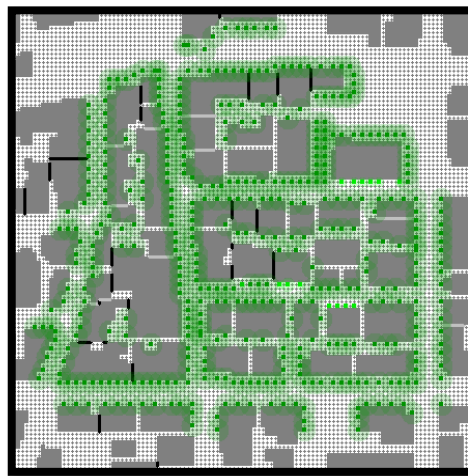


Figure 3. Urban vegetation model with the addition of 25% trees.

Thermal Comfort

Thermal comfort is referred to as an indicator to describe people's individual experiences of temperature in urban open spaces. It is also defined as 'the condition of mind which expresses satisfaction with the thermal environment' [9]. The assessment of the thermal comfort was based on both PET and PMV.

To calculate the PET and PMV model, information about the personal human parameters (Table 3) should be provided in addition to the clothing parameters. These are needed for the energy balance of the body, which enables assessment of the comfort in a surrounding environment. The total metabolic rate is known as the internal energy production by the body. However, the clothing insulation is the thermal resistance of the clothing and mainly 0.5 clo for summer clothes and 1.5 for winter clothes [9]. PET expands the human thermal comfort to a standard index with units of °C, while PMV is also used as thermal comfort index and ranges from very cold (-3.5) to very hot (3.5), as shown in Table 4.

Table 3. Setting for PET Calculation

Input	
Weight (Kg)	70
Height (m)	1.75
Surface Area (m ²)	1.9
Walking Speed (m/s)	1.2
Clothing Insulation (clo)	0.5
Total Metabolic Rate (W)	166.46

Table 4. Thermal Sensation According to PMV and PET Values

Grade of Physiological Stress	Thermal Perception	PET °C	PMV
Extreme Cold Stress	Very Cold	< 4	< -3.5
Strong Cold Stress	Cold	4 - 8	-3.5 - -2.5
Moderate Cold Stress	Cool	8 - 13	-2.5 - -1.5
Slight Cold Stress	Slightly Cool	13 - 18	-1.5 - -0.5
No Thermal Stress	Comfortable	18 - 23	-0.5 - 0.5
Slight Heat Stress	Slightly Warm	23 - 29	0.5 - 1.5
Moderate Heat Stress	Warm	29 - 35	1.5 - 2.5
Strong Heat Stress	Hot	35 - 41	2.5 - 3.5
Extreme Heat Stress	Very Hot	> 41	> 3.5

RESULTS AND DISCUSSION

Figure 4 shows the change of air temperature (T_a) and mean radiant temperature (T_{mrt}) on a summer day (20th August) at two different streets at human height level 1.4 m. The comparison is between the base case scenario and the vegetation scenario after adding 25% of urban vegetation coverage. At street S1, the T_a is reduced by 1.9 °C during the night-time, while the most significant reduction is observed at 12:00 pm and 17:00 by 2.5 °C. On the same street, the biggest T_{mrt} reduction was observed at noon by 19.8 °C. However, at street S2, the T_a is reduced by 2.7 °C at 9:00 am, and by 2.2 °C through the additional hours. For the T_{mrt} , the biggest reduction was observed at 9:00 am by 28.5 °C. The decrease in T_a and T_{mrt} during the daytime by adding the urban vegetation shows the immense advantage and importance of the shading that comes from the trees.

Adding vegetation in the proposed scenario will lead to shaded space, thus a decrease in SVF, which is better than the thermal environment in an open space with high value in SVF. Therefore, increasing the urban vegetation in any area will protect the pedestrian from direct sunshine due to the shading produced, resulting in decreasing the T_{mrt} and T_a and improving the thermal comfort in the summer days.

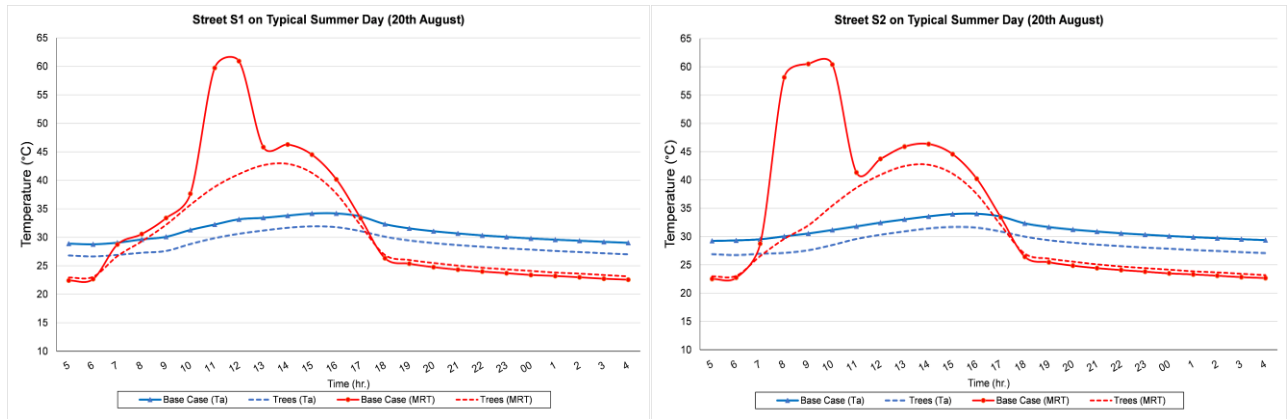


Figure 4. Average air temperature (T_a) and mean radiant temperature (T_{mrt}) at two streets with the base case scenario and the vegetation scenario at 1.4m above the ground level.

Figure 5 shows the thermal comfort change on a summer day (20th August) at the two streets at human height level 1.4 m. Adding the vegetation results in lower values of PET compared to the base case scenario at the two chosen streets. For street S1, the most significant reduction (12.6 °C) was observed at 12:00 pm. The average PET reduction on the same street was found to be 2.81 °C. However, for street S2, the most significant decrease after adding the trees was observed at 9:00 am by 21 °C, and the average PET reduction was found to be 3.8 °C. The PET increases with the increase of T_{mrt} and T_a [10]. The most significant improvement happened at 9:00 am, is because of the reduction occurred in the T_{mrt} for street 2, as shown in figure 4, with a difference of 28.5 °C. These results are attributable to the decrease in the solar radiation that reached the ground surface resulted from the shading of the trees.

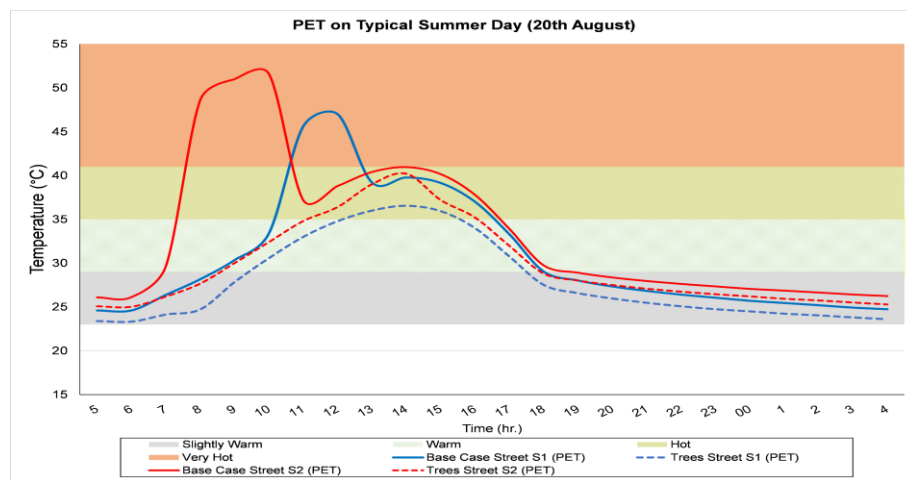


Figure 5. PET change at 1.4m above the ground level on a typical summer day (20th August) at two locations with the base case scenario and the vegetation scenario.

Figure 6 and figure 7 show the PET and PMV distribution map on a summer day at three different hours 8:00 am, 12:00 pm, and 16:00, respectively. The addition of trees and depending on the location of the sun results in lowering PET and PMV indices, as shown in table 4.

Besides, adding trees show that the vegetation scenario, has lower PET and PMV index (more comfortable) than the current base case scenario. Also, showing the figures of streets S1 and S2 with different PET and PMV values,

give an idea about which street to be pedestrianized during the daytime.

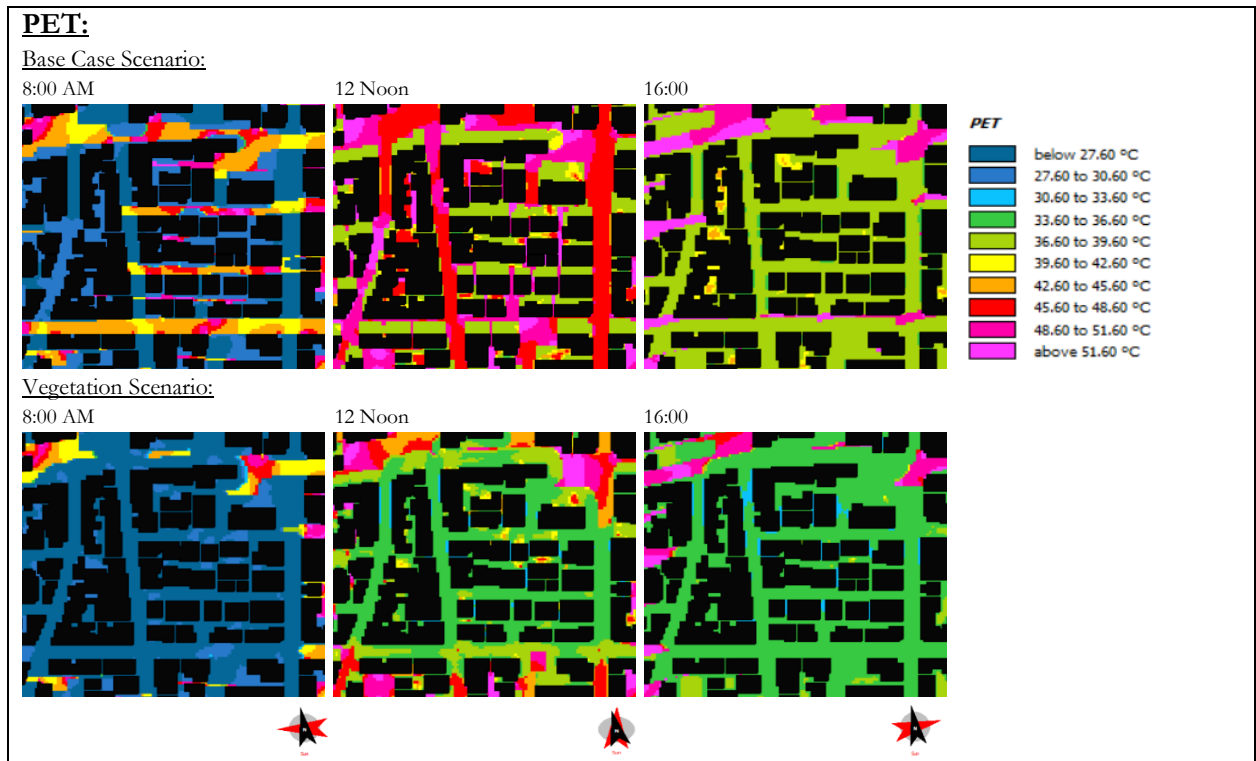


Figure 6. PET simulation for both scenarios at three different hours.

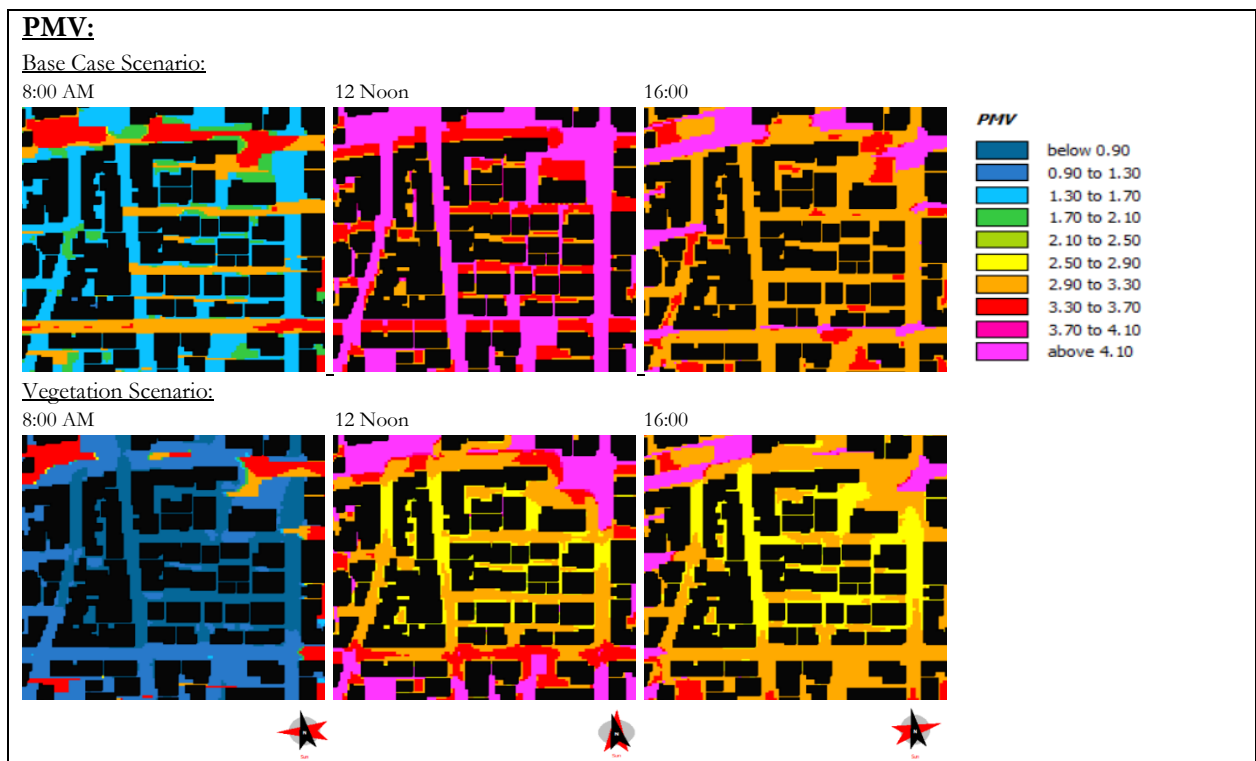


Figure 7. PMV simulation for both scenarios at three different hours.

CONCLUSION

Increasing the vegetation cover would help and protect the pedestrian from direct sunshine, thus reduce the mean radiant temperature and improve thermal comfort based on the results.

Compared with the two models, the base case scenario has higher PET values than the proposed model. The outdoor air temperature could be reduced by 2.5 °C during the daytime after adding urban vegetation. Also, a reduction in air temperature is observed at night. An average of 3.8 °C PET on a summer day could be reduced compared to the base case scenario. Nevertheless, the overall results in both models are not in the ideal comfort zone PET (18 - 23 °C) and PMV (-0.5 - 0.5), as most thermal perception has reached after adding the vegetation was slightly warm and warm during the early morning and nighttime.

Hopefully, these results will contribute to show the effect of the UHI mitigation technique for sustainable urban planning in Beirut. This study will identify the significant benefits that can result in increasing vegetation in urban areas and can help inform planning and design guidelines applicable to the city and buildings. The future-plan that was investigated needs to be reviewed and more mitigation strategies to be studied to reach the comfortable thermal comfort standard. Also, it is necessary to find ways in urban planning to mitigate the adverse effects of global warming in the Mediterranean cities, and urban vegetation should be considered as an additional guideline in mitigation strategies for outdoor spaces. Further investigation is needed to merge pollution dispersion in assessing mitigation strategies.

NOMENCLATURE

T_a = Air Temperature

T_{mrt} = Mean Radiant Temperature

PET = Physiological Equivalent Temperature

PMV = Predicted Mean Vote

SVF = Sky View Factor

α = Albedo

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